

SymPy Bug Card

Bug 26: solveset returns a false solution for $\operatorname{acot}(x) = -\pi/2$

Task. Solve the real inverse-cotangent equation.

$$\operatorname{acot}(x) = -\frac{\pi}{2}, \quad x \in \mathbb{R}.$$

SymPy's answer.

$$\operatorname{solveset}(\operatorname{acot}(x) = -\pi/2, x, \mathbb{R}) = \{0\}.$$

Correct answer.

$$\emptyset.$$

Explanation. Substituting SymPy's returned point gives $\operatorname{acot}(0) = \pi/2$, so the residual for $\operatorname{acot}(x) + \pi/2 = 0$ is π . The value $-\pi/2$ is outside SymPy's principal real range for acot , recorded as $(-\pi/2, \pi/2]$.

Likely root cause. The diagnosis identifies `sympy/solvers/solveset.py`'s generic real inversion path applying `acot.inverse()`, namely `cot`, to $-\pi/2$ without checking the principal branch range. This is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-026-solveset-returns-a-false-solution-for-acot-x-pi-2/](#)